

Architecture Specification v1.0

Parameterized INT8/FP16 Systolic Tensor Core with Online Softmax Unit
Target: Arty A7-100T (XC7A100T) — FCCM 2027 Reconfigurable Computing Challenge

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Project Goal

Design and implement a fully parameterized, synthesizable systolic tensor core on the Xilinx XC7A100T FPGA that accelerates INT8 matrix multiplication with FP16 accumulation and an online softmax unit, targeting the attention score computation $QK^\top$ in transformer self-attention. The design is submitted to the FCCM 2027 Reconfigurable Computing Challenge. The summer 2026 deliverable is a verified, timing-closed N=8 prototype. The April 2027 submission extends to N=16 via tiling. All RTL is parameterized with N as a **generate** parameter.

System Architecture

The top-level design is a four-stage feed-forward pipeline. BRAM ping-pong buffers supply Q and K tiles to a skew network, which feeds the $N \times N$ systolic array. Each PE output passes through a dedicated INT32→FP16 conversion unit, and the resulting FP16 tile streams into the online softmax unit. The systolic array computes $C = QK^\top$ tile by tile, with the softmax consuming each completed output tile in a single streaming pass.

This architecture is motivated by recent work on fusing FlashAttention within a single systolic array [1], which demonstrates that interleaving matrix multiplication and softmax on the same array is feasible and eliminates the external vector unit overhead incurred by conventional accelerators.

Decision 1 — Dataflow: Output-Stationary

The systolic array uses **output-stationary** dataflow. Weight-stationary is a poor fit for $QK^\top$ because K is not a static weight matrix, it is a runtime activation that changes every forward pass, so the amortization benefit that weight-stationary provides for linear layers does not apply here. Additionally, keeping partial sums local to each PE avoids FP16 rounding errors: floating-point addition is not associative, and moving partial sums out of a PE and back changes the order of operations, accumulating rounding error at each round-trip. Output-stationary eliminates this by pinning the running sum in one register for the entire inner-product computation.

Decision 2 — Array Size: N=8 Base-line, N=16 Target

The RTL is parameterized with N as a **generate** parameter. The summer prototype targets **N=8**: 64 PEs, 64 DSP48E1 slices consumed, 176 DSPs remaining. The April 2027 submission targets **N=16** via tiling: the 16×16 matrix is decomposed into four 8×8 tiles, computed sequentially on the same physical array, keeping peak DSP usage at 64 and avoiding the 256-slice budget violation a flat N=16 array would require on the XC7A100T's 240 available slices [2].

A flat N=8 array at 1 PE per DSP48E1 consumes 64 DSPs. The Digilent Arty A7-100T provides 240 DSP slices and 135 BRAM36 primitives [3]. N must be a power of 2 and the matrix dimension must be a multiple of N for the tiling schedule to be correct.

Table 1: Resource budget: N=8 prototype.

Resource	N=8	Available
DSP48E1	64	240
BRAM36	16	135
DSP remaining		176
BRAM remaining		119

Decision 3 — Precision: INT8×INT8 → INT32 → FP16

Input operands are signed INT8. The INT8×INT8 product is a signed INT16 value, sign-extended and accumulated inside the DSP48E1's 48-bit accumulator [2], treated as an INT32 accumulator for downstream purposes. For N=8, the worst-case accumulated sum is:

$$8 \times 128 \times 127 = 129,032 < 2^{18} \quad (1)$$

requiring 18 magnitude bits plus one sign bit (19 bits total), comfortably within INT32 and well within the 48-bit DSP accumulator with no overflow risk.

Once tile accumulation is complete, the INT32 result is forwarded to a dedicated LUT-based INT32→FP16 conversion unit outside the DSP. The DSP48E1 is a purely integer primitive and cannot perform this conversion internally [2]. FP16 format: 1 sign bit, 5 exponent bits ($e_{\max} = +15$, $e_{\min} = -14$), 10 explicit mantissa bits. The conversion maps a 19-bit integer significand to 10 mantissa bits, losing approximately 9 bits of precision.

This loss is acceptable for attention logit computation: softmax is dominated by relative differences between scores rather than absolute integer fidelity. Small absolute errors in QK^T do not materially shift the output probability distribution after exponentiation and normalization.

Rounding mode: Round-to-Nearest Ties-to-Even (RNTE). RNTE is the IEEE 754-2008 default rounding mode [4]. It minimizes statistical bias over repeated rounding events by resolving tie cases toward the even mantissa bit, as opposed to always rounding up, which introduces a systematic positive bias. RNTE is applied once, at the moment the full accumulated INT32 tile result is converted to FP16, minimizing the total number of rounding events in the pipeline.

Decision 4 — BRAM Budget: 16 BRAM36

Bandwidth is the binding constraint in all cases, not capacity. The systolic array consumes or produces 8 values per cycle (one per PE row or column). Using True Dual-Port BRAM36 at 32-bit data width (the widest standard configuration [2]):

Table 2: Ping-pong buffer BRAM breakdown.

Buffer	BW/cycle	BRAMs	Ping-pong
Q tile	64 bits	2	$\times 2 = 4$
K tile	64 bits	2	$\times 2 = 4$
Output	128 bits	4	$\times 2 = 8$
Total			16 BRAM36

Ping-pong buffering is required because DMA loading of the next tile and systolic array consumption of the current tile access the same logical address space simultaneously — a true read/write conflict on address 0...63 that True Dual-Port mode cannot resolve within a single buffer. With ping-pong, compute and memory transfer are fully overlapped with zero stall. 119 BRAM36 primitives remain for the softmax unit and control logic.

Decision 5 — Pipeline Depth and Tile Latency: 26 Cycles

The output validity time for PE at coordinate (i, j) in an $N \times N$ output-stationary systolic array is:

$$t_{i,j} = i + j + N \quad (2)$$

derived from the systolic wavefront propagation model introduced by Kung and Leiserson [5]. The worst-case PE at $(N-1, N-1)$ gives:

$$t_{\max} = (N-1) + (N-1) + N = 3N - 2 \quad (3)$$

For $N=8$: $3(8) - 2 = 22$ cycles of propagation latency. The DSP48E1 internal pipeline (input register → multiplier register → pre-adder register → accumulator register) adds a fixed **4-cycle** depth [2]. In output-stationary dataflow, PE outputs do not propagate to neighboring PEs, so this depth is a one-time additive offset, not a per-hop multiplier:

$$T_{\text{tile}} = (3N - 2) + T_{\text{DSP}} = 22 + 4 = \mathbf{26 \text{ cycles}} \quad (4)$$

Critical RTL requirement: Input data must be skewed by exactly 1 cycle per row index for Q inputs and 1 cycle per column index for K inputs at the array boundary. Without this skewing, PE (i, j) receives misaligned operands and the dot product is computed incorrectly.

Decision 6 — Online Softmax: AHBU exp() (TBD)

The online softmax uses the numerically stable single-pass formulation first described by Milakov and Gimelshein [6]:

$$\text{softmax}(X)_i = \frac{e^{x_i - m}}{\sum_{j=1}^N e^{x_j - m}}, \quad m = \max(X) \quad (5)$$

This limits $\exp()$ inputs to the domain $[-8, 0]$ and outputs to $(0, 1]$, preventing overflow. A running maximum m and running sum ℓ are maintained and updated in a single streaming pass over the FP16 output tile from the systolic array, following the rescaling update rule $\ell \leftarrow \ell \cdot e^{m_{\text{old}} - m_{\text{new}}} + e^{x_i - m_{\text{new}}}$ when a new maximum is encountered. This is the same algorithm that underpins FlashAttention [1].

The $\exp()$ function will be implemented using the **Approximate Hybrid Binary-Unary (AHBU)** method of Khataei, Singh, and Bazargan [7]. At 8-bit resolution, AHBU Config. 1 achieves mean absolute error below 7.07×10^{-3} using 4 LUTs and 1-cycle latency, outperforming CORDIC and piecewise-linear LUT by multiple orders of magnitude in area $\times$ latency (Table 3 of [7]). The down-scaling formulation of Eq. 5 restricts the $\exp()$ input to $[-8, 0]$, allowing the AHBU unit to operate at fixed-point precision Fixed<1, 16, 12> as used in the reference paper.

Outstanding items for May 7 meeting with Prof. Bazargan:

- AHBU (FPGA '23) vs. CompressedLUT (FPGA '24): contribution boundary for competition.
- Config. 1 vs. Config. 2 accuracy for attention softmax.
- Fixed<1, 16, 12> compatibility with FP16 systolic output.
- Running-max and running-sum BRAM cost.

Decision 7 — Clock Target: 200 MHz Conservative

The DSP48E1 standalone $F_{\max}$ on 7-series is 600 MHz per AMD UG479 [2]. This figure applies to the primitive in isolation with fully registered ports and no fabric routing. The realistic achievable $F_{\max}$ for the full design on Artix-7 speed grade -1 is lower due to three candidate critical paths:

1. **Input skew logic:** shift register chains feeding each PE row and column introduce fabric routing delay of approximately 1–2 ns per hop on Artix-7.
2. **INT32→FP16 conversion LUT chain:** combinational depth of the rounding and normalization logic not yet characterized.
3. **Online softmax exp() unit:** AHBU combinational depth is a function of configuration parameters (IL, $L_{\min}$, K, TRE) and has not yet been synthesized for this target device [7].

The systolic array alone, with fully pipelined DSP48E1 primitives and registered inter-PE connections, is expected to close timing at **450 MHz**. The full design clock target is set conservatively at **200 MHz** pending softmax characterization after the May 7 Bazargan meeting. The target will be revised upward once the softmax critical path is measured in Vivado synthesis.

Open Items — May 7, 2026

- AHBU vs. CompressedLUT: FCCM contribution boundary.
- Config. 1 vs. 2 accuracy for attention logits.
- Fixed $\langle 1, 16, 12 \rangle \leftrightarrow$ FP16 conversion stage.
- Online softmax BRAM cost (running max + sum registers).
- Softmax $F_{\max}$ on Artix-7: update clock target post-synthesis.

This document is committed to GitHub as the architectural baseline. No RTL is written prior to this lock.

References

- [1] J. Lin, Y. Li, G. Chen, and T. Bourgeat, “SystolicAttention: Fusing FlashAttention within a single systolic array,” 2025.
- [2] AMD (Xilinx), “7 series DSP48E1 slice user guide,” Tech. Rep. UG479 v1.10, Advanced Micro Devices, Inc., 2018.
- [3] Digilent, “Arty A7-100T: Artix-7 FPGA development board,” 2024.
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- [5] H. T. Kung and C. E. Leiserson, “Systolic arrays (for VLSI),” in *Sparse Matrix Proceedings 1978*, (Philadelphia, PA), pp. 256–282, Society for Industrial and Applied Mathematics, 1979.
- [6] M. Milakov and N. Gimelshein, “Online normalizer calculation for softmax,” *CoRR*, vol. abs/1805.02867, 2018.
- [7] A. Khataei, G. Singh, and K. Bazargan, “Approximate hybrid binary-unary computing with applications in BERT language model and image processing,” pp. 1–11, 2023.